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BCS-3C
LAB 6

Q1: Write a program that Count the number of one bits in DX and its complement and check either the both count equal or not.

org 0x100

jmp start

num: dw 8 ;0103 (the address of labels)
count1: dw 0 ;0105,,,to store original number 1's count
count2: dw 0 ;0107,,,to store compliment number 1's count

2	3	4	5	6	7
DS:0103	08	00	01	00	0F

count:

ror ax,1
jnc skip
inc bx

skip:

dec cx
cmp cx,0
jne count
ret

start:

mov ax, [num]
mov cx, 16
mov bx, 0
call count
mov [count1], bx

mov ax, [num]
not ax
mov cx, 16
mov bx, 0
call count
mov [count2], bx

mov ax, 0x4c00
int 0x21

DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program:...

AX	SI	CS	IP	Stack	Flags
0000	0000	19F5	0100	+0 0000	7202
BX	DI	0000	DS	+2 20CD	
0000	0000	19F5		+4 9FFF	OF DF IF SF ZF AF PF CF
CX	BP	0000	ES	+6 EA00	0 0 1 0 0 0 0
0000	SP	FFFE	SS		
			FS		
			19F5		

CMD >SS

0100	E91300	JMP	0116
0103	0800	OR	[BX+SI],AL
0105	0100	ADD	[BX+SI],AX
0107	0F00D1	LLDT	CX
010A	C8730143	ENTER	0173,43
010E	49	DEC	CX
010F	81F90000	CMP	CX,0000
0113	75F4	JNZ	0109

2	3	4	5	6	7	8	9	A	B	C	D	E	F	0	1	2
DS:0103	08	00	01	00	0F	00	D1	C8	73	01	43	49	81	F9	00	00
DS:0113	75	F4	C3	A1	03	01	B9	10	00	BB	00	00	E8	E7	FF	89
DS:0123	1E	05	01	A1	03	01	F7	D0	B9	10	00	BB	00	00	E8	D5
DS:0133	FF	89	1E	07	01	B8	00	4C	CD	21	46	DC	00	00	8E	5E
DS:0143	FC	83	7D	0E	00	74	09	8B	46	F2	48	3B	46	F6	7E	08

1 Step 2ProcStep 3Retrieve 4Help ON 5BRK Menu 6 7 up 8 dn 9 le 10 ri

Q2. Write a Subroutine that will allocate grades according to the marks entered as below. (Setup a memory variable for marks and grade)

Marks Grade

90-100 AA

80-89 AA

70-79 BB

60-69 CC

50-59 DD

40-49 EE

<=39 FF

org 0x100

jmp start

marks: dw 95,85,72,60,55,40,30

grade: db 0,0,0,0,0,0

grading:

cmp ax, 80

jge aa

cmp ax, 70 ;line 10

jge bb

cmp ax, 60

jge cc

cmp ax, 50

jge ddg ;line 15

cmp ax, 40

jge ee

cmp ax, 39

jle ff

aa:

mov byte [grade+si],0xAA

ret

bb:

mov byte [grade+si],0xBB

ret

cc:

mov byte [grade+si],0xCC

ret

ddg: ;line 30

mov byte [grade+si],0xDD ;line 31

ret

ee:

mov byte [grade+si],0xEE

ret

ff:

mov byte [grade+si],0xFF

ret

start: ;line 40

mov bx, 0

mov cx, 7

mov si, 0

l1: ;line 46

```
mov ax, [marks+bx]
call grading
inc si
add bx,2
dec cx
cmp cx, 0
jnz l1 ;line 54
```

```
mov ax,0x4c00
int 0x21
```

DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program:...

AX	SI	CS	IP	Stack	Flags
0000	0000	19F5	0100	+0 0000	7202
BX	0000	DI	0000	+2 20CD	
CX	0000	BP	0000	+4 9FFF	OF DF IF SF ZF AF PF CF
DX	0000	SP	FFFE	+6 EA00	0 0 1 0 0 0 0 0

CMD >

Program terminated OK

0100	E95600	JMP	0159
0103	5F	POP	DI
0104	005500	ADD	[DI+00],DL
0107	48	DEC	AX
0108	003C	ADD	[SI],BH
010A	0037	ADD	[BX],DH
010C	0028	ADD	[BX+SI],CH
010E	001E00AA	ADD	[AA00],BL

1	2	3	4	5	6	7	8
DS:0111	AA	AA	BB	CC	DD	EE	FF
DS:0119	00	7D	19	3D	46	00	7D
DS:0121	3D	3C	00	7D	1B	3D	32
DS:0129	7D	1C	3D	28	00	7D	1D
DS:0131	27	00	7E	1E	C6	84	11
DS:0139	AA	C3	C6	84	11	01	BB
DS:0141	C6	84	11	01	CC	C3	C6
DS:0149	11	01	DD	C3	C6	84	11
DS:0151	EE	C3	C6	84	11	01	FF
DS:0159	BB	00	00	B9	07	00	BE

2	3	4	5	6	7	8	9	A	B	C	D	E	F	0	1	2
DS:0103	5F	00	55	00	48	00	3C	00	37	00	28	00	1E	00	AA	AA
DS:0113	BB	CC	DD	EE	FF	50	00	7D	19	3D	46	00	7D	1A	3D	3C
DS:0123	00	7D	1B	3D	32	00	7D	1C	3D	28	00	7D	1D	3D	27	00
DS:0133	7E	1E	C6	84	11	01	AA	C3	C6	84	11	01	BB	C3	C6	84
DS:0143	11	01	CC	C3	C6	84	11	01	DD	C3	C6	84	11	01	EE	C3

1 Step 2ProcStep 3Retrieve 4Help ON 5BRK Menu 6 7 up 8 dn 9 le 10 ri

Marks label starts from 0103 to 010F in m2

2	3	4	5	6	7	8	9	A	B	C	D	E	F
DS:0103	5F	00	55	00	48	00	3C	00	37	00	28	00	1E

Grade label starts from 0111 to 0117 in m1 with grade of every mark in corresponding index

1	2	3	4	5	6	7
DS:0111	AA	AA	BB	CC	DD	EE